

Introduction to Quantum Report

Quantum Computing

Fundamentally different approach of computing that uses quantum mechanical phenomena for computing process. They use qubits instead of traditional bits for processing instead of classical bits of 0 and 1.

What Advantages does it have over traditional classical computing.

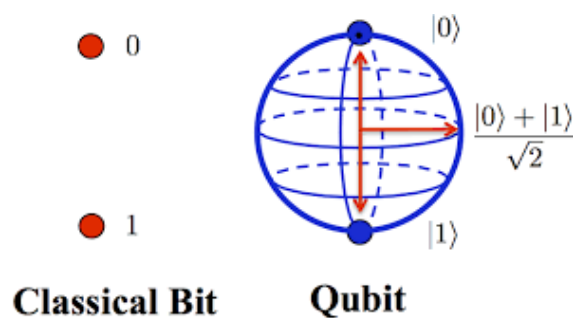
- Better at exponential scaling problems
Some problems grow in complexity based on the increase in complexity. Example "Simulating the behaviour of atoms in a molecule". The lack of state in classical computing is covered up by the many qubit states of quantum computing .
- Trying many possibilities at once
Traditional computing checks solutions sequentially ie, $A \rightarrow B \rightarrow C$ and D . Quantum Computers can explore multiple paths simultaneously through superposition. For problem with vast encryption or delivery optimization.

Classical VS Quantum Computing

Classical Computing	Quantum Computing
Uses bits 1 and 0	Uses qubits of multiple bits
Processes sequential / limited parallel processing ability	Massively parallel through superposition
Deterministic Output [same input = same output]	Probabilistic output [same input = probability distribution of outputs]
Uses Logical Gates : AND , OR , NOT	Uses : Hadamard , CNOT, Pauli
Bits are independent	Qbits can be entangled .
Reading doesn't change the bits	Measuring collapses superposition, destroys quantum state.

Classical Computing	Quantum Computing
Programmed step by step following instructions	Uses Quantum Circuits , probability amplitudes ,interference patterns
Used for general computing ,database web and so on	Used for Quantum simulation ,cryptography, optimization, drug discovery
Uses algorithm like : Sorting , searching, encryption (RSA)	Uses Algorithm like : Shor's algorithm , Grover's algorithm , VQE

Bits vs Qubits



Classical Bits

Having two definite states 0 and 1 and no states in between .

Various ways to define multiple states

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OFF-OFF-OFF [000]
OFF-ON-ON [011]
and so on
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Possible patterns for the number of bits can be expressed as 2^N . There can only be one pattern at any moment and to see each state ,each state must be computed at a time individually .

Quantum Bits (QUBITs)

A qubit is the fundamental unit of quantum information that can exist in a superposition of states . Unlike a classical bit that is either 0 or 1 , a qubit can exist in both states simultaneously through superposition.

Mathematical Definition

A qubit state is represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where:

- $|\psi\rangle$ = qubit state (using Dirac "ket" notation)
- $|0\rangle$ = computational basis state 0
- $|1\rangle$ = computational basis state 1
- α, β = complex probability amplitudes

Constraint:

$$|\alpha|^2 + |\beta|^2 = 1 \text{ (normalization)}$$

Qubits can be any state within 0 and 1 where α, β being the probability of 0 and 1, the value collapses to either 0 or 1 after it is measured.

Comparison Table

Bits	QuBITs
Smallest bits 0 and 1	Smallest unit of quantum ,superposition of 0,1 or both
Discrete and Deterministic	Continuous probability amplitudes
Uses Boolean logic	Follows complex vectors and linear algebra
measurement result always known before measurement	Measurement Result is probabilistic which collapses on measurement
Concepts like superposition and entanglement are not possible	Superposition and Entanglement are heavily used
very low error sensitivity and is stable	Extremely fragile and high error sensitivity.
n bits are stored in 2^n values	n bits represent all 2^n states simultaneously.
Measurement doesn't destroy quantum states	Measurement destroys quantum state

Bloch Sphere

Geometric representation of qubits as points on a sphere.

- Only works for qubits (single qubits system)
- Cannot extend to 3+ classical states.
- Also called "unit 2-sphere" (surface is 2 dimensional)

Mathematical Foundation

General qubit state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Spherical coordinate form :

$$|\psi\rangle = \cos(\theta/2)|0\rangle + (e^{i\phi}) * \sin(\theta/2)|1\rangle$$

where

- $\theta \in [0, \pi]$ = polar angle
- $\phi \in [0, 2\pi)$ = azimuthal angle
- $\alpha \geq 0$ (can always choose this without loss of generality due to global phase)

Parameter uniqueness:

- If $\beta = 0 \rightarrow \theta = 0$, ϕ arbitrary (north pole)
- If $\alpha = 0 \rightarrow \theta = \pi$, ϕ arbitrary (south pole)
- Otherwise $\rightarrow$ unique (θ, ϕ) pair

Deriving density matrix

$$|\psi\rangle\langle\psi| = \begin{pmatrix} \cos^2(\theta/2) & e^{-i\phi} \cos(\theta/2) \sin(\theta/2) \\ e^{i\phi} \cos(\theta/2) \sin(\theta/2) & \sin^2(\theta/2) \end{pmatrix}$$

from [IBM quantum cloud](#)

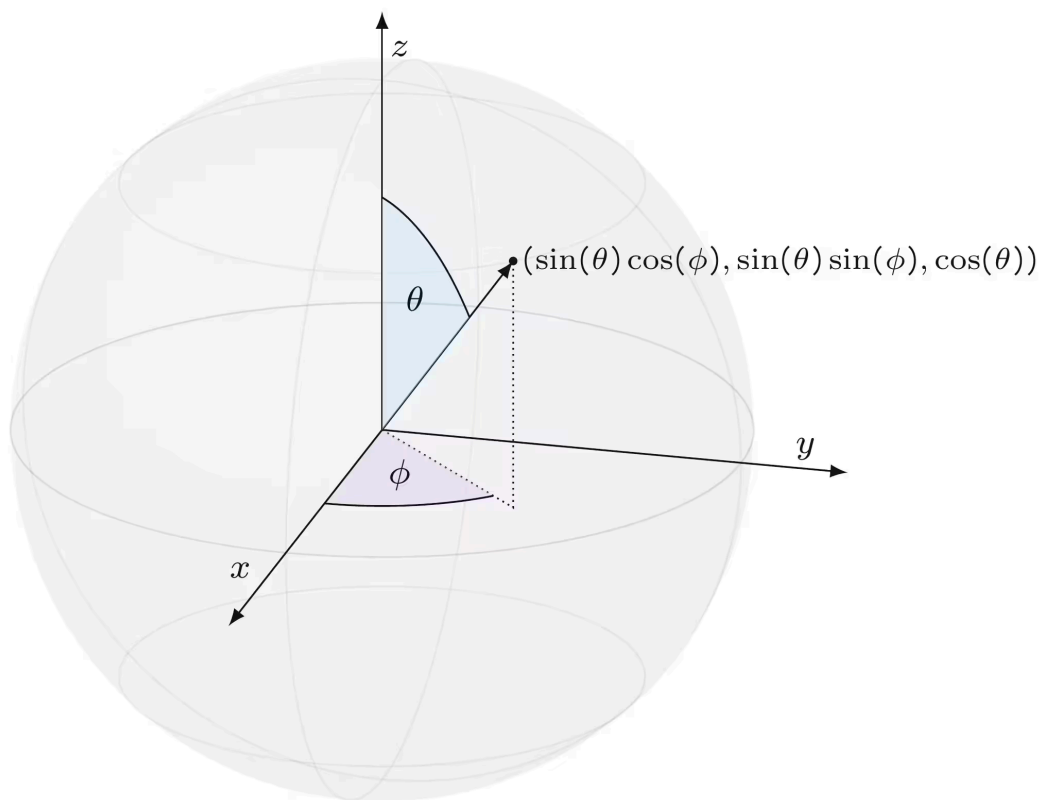
Upon computing the matrices we get a identity matrix :

$$(\sin(\theta)\cos(\phi), \sin(\theta)\sin(\phi), \cos(\theta))$$

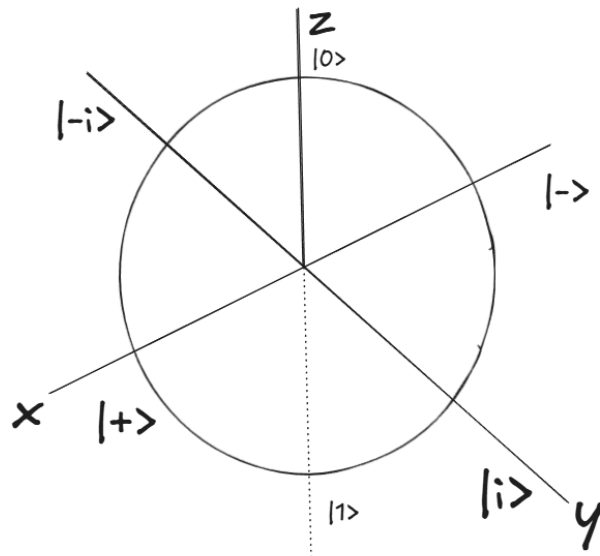
Where :

Spherical coordinates: $(r, \theta, \varphi) = (1, \theta, \varphi)$

- $r = 1$ (always unit radius for pure states)
- $\theta =$ polar angle (0° to 180°)
- $\varphi =$ azimuthal angle (0° to 360°)



Let us consider a self made diagram



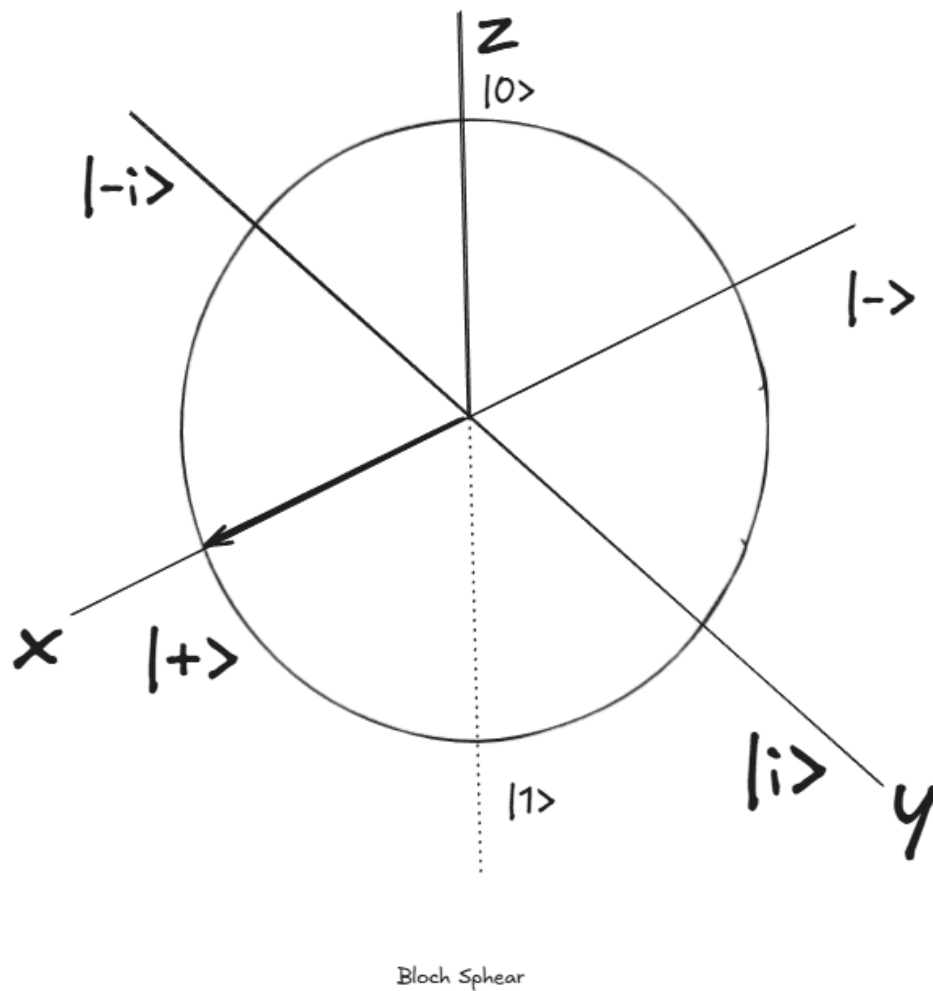
Bloch Sphere

Suppose we have the qubit value

$$|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

the the probable position will be :

squaring both the numbers we get 1/2 and 1/2 which gives us exactly half way between $|0\rangle$ and $|1\rangle$.

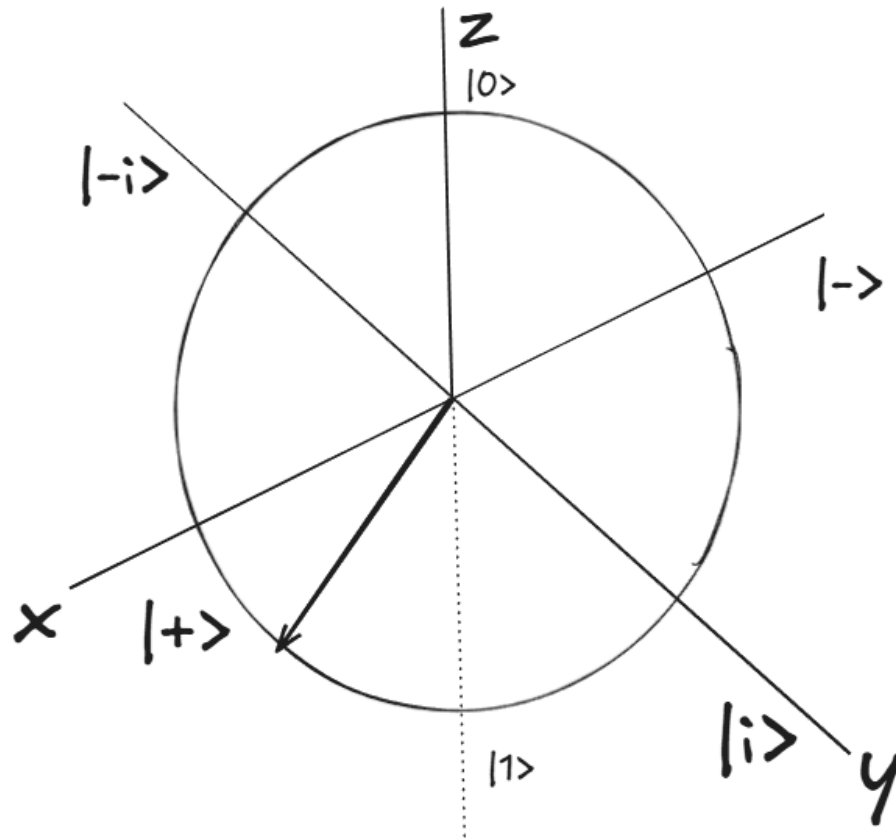


because it has even chance of being 0 or 1, since it is a sphere ,the qubit can spin around the sphere at that point ,which is called phase angle or simply phase.

Let us consider another example ,

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

taking square of both alpha and beta , we see 1/4th of $|0\rangle$ and 3/4th of $|1\rangle$,this mean there is a higher chance of 1 ,so the qubit lies somewhere around here .



Bloch Sphear

Quantum Entanglement

What is Quantum Entanglement?

Quantum entanglement is a fundamental phenomenon in quantum physics where two or more particles become linked in such a way that the state of one particle instantly determines the state of the other, no matter how far apart they are.

Entangled State in Quantum Computing

An entangled state in quantum computing is a special type of multi-qubit state in which the individual qubits do not have independent values. Instead, their states are linked : measuring one qubit instantly reveals information about the others.

A classic example is Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This means if qubit A goes to 0 , qubit B will go to 0 , and similarly with 1 as well , before measured ,neither qubit individually has a definite value.

Many famous quantum algorithms rely heavily on entanglement:

- Shor's Algorithm
- Grover's Algorithm
- Quantum Simulation

Quantum Error Correction

Error correction in quantum computing:

- Uses entanglement to spread information across many qubits
- Let's you detect errors without directly measuring the stored quantum information

Operation on PennyLane

